

THE PLACE OF THE FAST BREEDER REACTOR IN THE REACTOR DEVELOPMENT PROGRAMME IN JAPAN

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1. Introduction

In giving an account of the place of the fast breeder reactor in Japan's reactor development programme, the significance of her nuclear power generation and her need for the fast breeder reactor is indicated by explaining the nature of the power situation. The position of her nuclear power reactor development is given, with reference to the construction of the proven type reactors, the graphite moderated gas cooled reactor and the light water moderated and cooled reactor, which are at present contemplated, and with reference also to the development of the fast breeder reactor in the future. Finally, an account is given of plans for research and development of the fast breeder reactor in Japan.

2. Power Situation and Necessity of Developing the Fast Breeder Reactor in Japan

Long-range Prospect of Power Demand and Supply and the Significance of Nuclear Power Generation

Japan's economy has made rapid progress, as indicated by the average annual increase of 9.9% in the gross national product for the period from 1953 to 1963. According to the Government Economic Council, Japan's economy will maintain a fairly high rate of growth compared with Western countries, although its tempo will be a little slower than in the past. The gross national product in 1968, 1975 and 1985 may be respectively about 1.5, 2.5, and 4.5 times that of 1963. In other words, it is estimated that the average annual rate of increase will be 8.1% from 1963 to 1968, 7.6% from 1968 to 1975, and 6.7% from 1975 to 1985 (see Table 1).

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TABLE 1. ECONOMIC INDICES

Item	Unit	Year				Average annual rate of increase (%)		
		1963 (actual)	1968	1975	1985	1963/1968	1968/1975	1975/1985
Gross national product	Thousand million yen	19,767	29,210	48,880	88,400	8.1	7.6	6.1
Mining manufacturing pro- duction index	100 for 1950	149.5	239.3	428	804	9.9	8.7	6.5
Total population	Million	96	101	109	116	1.0	1.0	0.7
Demand for energy	Million tons coal equivalent (7000 kcal/kg)	179	286	492	880	9.9	8.1	6.0

TABLE 2. PROSPECT OF DEMAND FOR POWER
(million tons 7000 kcal/kg coal equivalent)

Item	Year				Ratio (%)				Average annual rate of increase (%)			
	1963 (actual)	1968	1975	1985	1963	1968	1975	1985	1958/ 1963	1963/ 1968	1968/ 1975	1975/ 1985
Coal	48	50	53	54	26.9	17.5	10.9	6.1	3.5	0.8	0.9	0.1
Oil	77	153	297	561	43.2	53.5	60.3	63.8	27.6	14.7	9.9	6.6
Electricity	49	81	143	280	27.5	28.2	29.1	31.8	14.2	10.4	8.6	6.9
Others	7	8	10	11	4.1	2.7	2.0	1.2	0.1	0.5	3.8	0.9
Total*	179	286	492	880	100	100	100	100	13.7	9.9	8.0	6.0
Imported energy among the primary supply of energy	114	223	428	844	58.4	71.3	81.2	88.9	29.4	14.4	9.7	7.0

* Duplicated portion of consumption due to use of fuel for electricity for private use is excluded.

Keeping pace with this economic growth, the demand for energy has also been rapidly increasing. The total demand of 96 million tons (7000 kcal/kg coal equivalent is the basis throughout) in 1958 reached 179 million tons in 1963, and is estimated to increase to 286, 492, and 880 million tons respectively in 1968, 1975, and 1985.

Demand for oil and electricity has increased whilst demand for coal remains roughly at the same level. As seen in Table 2 the recent change in the structure of demand and supply of energy is noteworthy.

Japan has to depend on imports to meet almost the whole of the increasing demand for power, as she has few energy resources. Overseas supplies increased from 26% of the total consumption in 1958 to 58% in 1963, and will increase to 71% in 1968, 81% in 1975, and as much as 89% in 1985. In 1968, \$2230 million will have to be spent on some form of energy, accounting for 21% of the total sum spent by Japan on imports.

The principal form of imported energy is oil. Japan imported 19 million kilolitres of oil in 1958 which increased to 71 million kilolitres in 1963 and is estimated to reach 143 million in 1968. The proportion of oil in the supply of primary energy is increasing and will account for 86% in 1985, and the amount of imported oil will then be 575 million kilolitres, if Japan fails to realize nuclear power generation. This emphasizes the significance of nuclear power generation in the Government's co-ordinated energy policy.

The electricity industry of Japan developed chiefly with hydroelectric power. Recent power plants, however, are for the most part thermal owing to improvements in the economy of oil burning and the decrease in the number of sites suitable for hydroelectric power generation. Consequently the position of hydroelectric versus thermal power plants was reversed in 1962, and the proportion of thermal plants in the total electric power output has since been increasing, being 55% in 1963 and estimated to be 69% in 1968. The new thermal power plants mainly burn heavy oil and it is intended to replace them by nuclear power plants in the future.

The keynote of a co-ordinated energy policy in Japan is cheapness and assured supply, which nuclear power generation embodies. Furthermore, compared with oil fuel, nuclear power does not require as much foreign currency and thus improves the nation's balance of trade. Therefore Japan, being dependent largely on imported energy, expects much from the development of nuclear power. The capital and generating cost of nuclear power plant being constructed in Japan today is still higher than that of oil burning plant, but it will gradually be reduced by improved and advanced techniques. It is thought that nuclear power will be competitive with or economically better than the oil fuel generation after 1970, and at the latest by 1975. With regard to security of supply of nuclear fuel Japan also relies much on imports. However, when compared with oil, a small quantity of nuclear fuel produces a great deal of energy, and is easier to transport and store—at

about one-fifth of the cost of heavy oil. It is estimated that the amount of foreign currency necessary to operate a nuclear power plant for 20 years will be some 60% of the amount required for a heavy oil fuel plant if a light water reactor is used, and 40% or even 20% if a heavy water reactor is used.

If and when the fast breeder reactor is put into use, various energy supply problems in Japan should be solved.

Nuclear Power Generation and its Fuel Requirements

In the long-range programme on development and utilization of atomic energy set up in 1961, the Atomic Energy Commission referred to the scale

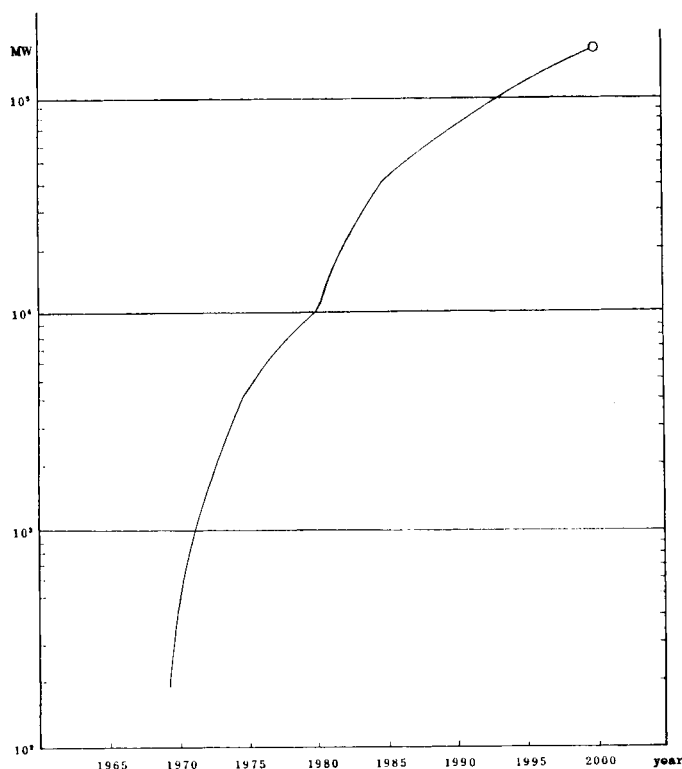


FIG. 1. Estimate of capacity of nuclear power generation.

of development of nuclear power generation, estimating that the economics of nuclear power generation will be comparable with steam power generation around 1970, with the expectation that about 1 million kW of nuclear power would be installed in the early stages up to 1970 and a further 6–8.5 million kW by 1980.

In view of this, the electricity industry has begun to plan nuclear power

plants. Plans released in 1965 gave a figure of 4.6 million kW as the installation expected by 1975, thus exceeding the figure proposed in the long-range programme of the Atomic Energy Commission.

The Japan Atomic Industrial Forum published in January 1966 the result of its work on the prospect of nuclear power generation up to the year 2000.

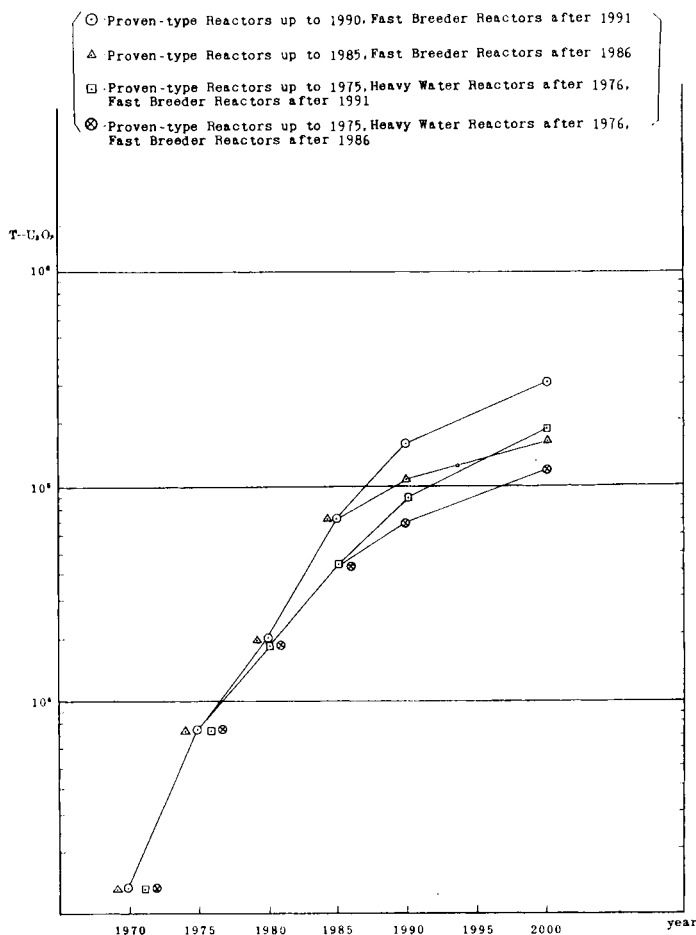


FIG. 2. Nuclear fuel demand (natural uranium equivalent).

The Forum estimates the output from nuclear power to be 4.8 million kW (10% of the total amount of electricity to be generated), 37.9 million kW (48%) and 121.7 million kW (63%), respectively for the periods 1965–75, 1975–85 and 1985–2000. At the end of the year 2000, nuclear power plants will have a capacity of 164 million kW, accounting for 47% of the total power plants and covering 66% (1164×1000 million kWh) of the demand in the year, according to the estimate by the Forum (Fig. 1).

With regard to the demand for nuclear fuel the Atomic Energy Bureau has worked out tentative figures on the following assumptions:

- (i) proven type reactors to be developed up to 1990, and fast breeder reactors to be developed after 1991;
- (ii) proven type reactors to be developed up to 1985, and fast breeder reactors to be developed after 1986;
- (iii) proven type reactors to be developed up to 1975, heavy water reactors to be developed after 1976, and fast breeder reactors to be developed after 1991;
- (iv) proven type reactors to be developed up to 1975, heavy water reactors to be developed after 1976, and fast breeder reactors to be developed after 1986.

The result of the calculation is shown in Fig. 2. In the case of (i) where the largest amount of uranium is required, the demand for nuclear fuel up to the year 2000 is 294,000 tons in terms of natural uranium. In the case of (iv), the nuclear fuel needed is 119,000 tons, i.e. 40% of the requirements of (i).

In this calculation, the proven type reactor is assumed to be light water reactor (BWR 50%, and PWR 50%) and the heavy water reactor is assumed to be natural uranium heavy water moderated, boiling light water cooled reactor. The depleted uranium is assumed to be recycled.

Uranium deposits in Japan are at present estimated to be about 4000 tons. They are so small that the big demand for uranium must be met by imports. Considering the present estimates of world uranium resources, early establishment of fast breeder reactors is to be hoped for, in view of security of energy supplies in Japan.

3. Basic Stand for Power Reactor Development and Plan for Development of Fast Breeder Reactor

The Atomic Energy Circle of Japan celebrated the tenth anniversary of the nation's atomic energy development in October 1965. In 1955 the Japanese Government laid down the Atomic Energy Basic Law and related laws and regulations to start research and development for the peaceful uses of atomic energy. In those days Japan had neither proper facilities for the development of atomic energy nor specialists to take charge of the research and development. The Atomic Energy Commission first proposed a policy centred on the training of competent men and the establishment of research establishments thus facilitating fundamental research. In the intervening 10 years, both the Government organization for atomic energy development and private industry have built up a sound technological foundation.

The Atomic Energy Commission now considers it time to start the development of power reactors. The Commission has been holding a Power Reactor

Development Conference of specialists in various fields, and has been making studies for a power reactor development programme directed towards the solution of the energy problems of Japan.

The Commission will before long reach a final conclusion and set up the power reactor development programme. Earlier, in July 1965, the Commission disclosed its basic ideas on how the development of power reactors should proceed. According to the Commission's ideas, the power reactors to be adopted will be the proven type, the advanced converter reactor and the fast breeder reactor.

The power reactors to be constructed in Japan for some time to come will be the proven type reactors and it is necessary to realize their domestic production as early as possible. However, since this type of reactor has already proved practicable in other countries, domestic production and improvements will be accomplished by employing foreign techniques. The Government must also take up, as a subsidy, research on safety problems and on fuels, including the utilization of plutonium. At the same time, the aim of long-term development will be the fast breeder reactor which will be in service about 20 years from now. The proven type reactors to be adopted in Japan will most likely be the reactors of the enriched uranium series. Therefore, considering the nuclear fuel situation in Japan, it is desirable that the advanced converter reactor which has a good neutron economy should be included in Japan's power generating scheme.

It is essential that the advanced converter reactor (1) has a better economy than the proven type reactors; (2) can make effective use of nuclear fuel, and (3) can pave the way for diversified use of nuclear fuel. Among several reactors having these characteristic features, a suitable type is to be picked out as the interim target of development.

On this basis, working groups were organized to study the development of the advanced converter reactor and the fast breeder reactor. As a result of their work, either the heavy water moderated, boiling light water cooled reactor, or the heavy water moderated, gas cooled reactor is likely to be selected for development as the advanced converter reactor.

As regards the fast breeder reactor, work is being undertaken by a group headed by Professor Oyama of Tokyo University. This working group submitted in January 1966 a report on proposals for the development of the fast breeder reactor in Japan. Subject to some modification and co-ordination it is due for formal adoption by the Atomic Energy Commission as the Government's development project.

The following passages indicate the plan for development of the fast breeder reactor in Japan, using the report as a basis. Study of the fast breeder reactor in Japan had already been started, though on a minor scale, when research started on atomic energy, particularly on reactor physics and sodium technology. Several years ago construction began of research facilities on a major

scale, such as a critical facility and a plutonium research facility. However, for the development of the fast breeder reactor, it is necessary for the Government to play a principal role and promote it as a national project, because the development of fast breeder reactors covers many branches of study, and requires considerable funds and time. It is highly desirable for Japan, with her scant energy resources and an extremely large demand for energy, to expedite the practical uses of the fast breeder reactor, to derive benefit from it, and to be thus qualified to maintain her position as an industrial country.

The plan assumes that the fast breeder reactor will be in commercial use in the nation's power system in the latter part of the 1980's, and envisages two stages in the course of development leading to its practical use. The first stage will be the period up to the mid-1970's when the prototype reactor comes on power. During this period all research activity will be concentrated on the construction and operation of a prototype reactor with an output of about 200 MW. In the first half of this first stage, an experimental reactor will be constructed, prior to the prototype reactor, to serve for a long time as a fuel irradiation facility and also to render a substantial contribution to the technical development of the prototype reactor. The second stage will be the demonstration reactor, i.e. the target will be starting the operation of a demonstration reactor of about 1000 MW in the early 1980's.

In the first stage of development, certain important items of research and development are called for prior to the construction of prototype reactor, such as the fundamental technology for the fast breeder reactor in the future, i.e. safety technology, cooling system technology, and fuel technology.

The main subject of safety technology is the reactor physics of the large core. With a view to facilitating the study of reactor physics, Japan Atomic Energy Research Institute (JAERI) has since 1963 been building a split type critical facility expandable to 3000 litres in core volume, with a budget of about 1000 million yen (£1 million). This facility will go critical with enriched uranium this year, and will be used with a plutonium core after 1969 to clarify the core characteristics of a large reactor, and at the same time, to carry out the core mockup experiment for the prototype and demonstration reactors. For engineering safety research, it is considered necessary to carry out experimental study to clarify the transient behaviour of cooling system, fuel meltdown, and so on.

For the coolant of the fast breeder reactor, sodium, steam, helium gas, etc., are possible. Compared with the sodium cooled reactor, the steam or gas cooled reactor has many technical problems yet to be solved. For this reason, Japan, which started work later than other countries, has to concentrate the primary part of the development activities on the sodium cooling, limiting the study on steam or gas cooling to basic research and investigation. Since 1957 sodium technology has been studied by Hitachi Ltd., and by JAERI with regard to various fundamentals, and some of the equipment for this

system has already been developed. In the future development of technology concerning the sodium cooling system, two devices are likely to be important. One is the 2 MW systems engineering loop. It is being designed on the basis of present experience, and will be constructed in 1967. The details of this loop appear in session 5B. It will enable Japan to carry out the various tests necessary for the experimental reactor. The second is the experimental reactor. Experience with the experimental reactor built before the prototype reactor is expected to play a big role in the development of prototype and demonstration reactors.

The plutonium fuel to be used in the fast breeder reactor is currently under study at JAERI and the Atomic Fuel Corporation (AFC). AFC is in charge of the study of manufacturing, whilst JAERI is engaged in basic research. At AFC, a plutonium laboratory provided with a production line was completed in 1965 with an appropriation of 1200 million yen (about £1.2 million), and study is now in progress on the coprecipitated pelletization method, the manufacture of high density UO_2/PuO_2 by Sol-Gel method, and on the fabrication method using the vibratory compaction process. The development of fuel for the fast breeder reactor calls for the irradiation test under fast neutron flux. For this purpose it is planned to construct an experimental reactor at the earliest possible date and conduct such irradiation tests for the fuel of the prototype reactor. However, the irradiation tests may be entrusted to overseas facilities before the completion of the experimental reactor.

JAERI is working on the preliminary design of the experimental reactor with a thermal output of 100 MW. In the design, attention is being paid to the mechanisms so as to enable them to be adapted following the results of the fuel irradiation test. The results of the various development tests will be incorporated in the design, and construction will begin in 1968. The experimental reactor is scheduled to be critical in 1972.

The development of the fast breeder reactor calls for research activities extending over many fields, and it entails large expenditure. Therefore, it is imperative for a country, such as Japan, where work was started late, to advance her research and development through international co-operation. Her concept of international co-operation is in two categories. The first is the exchange of information, and the second participation in co-operative development projects. As regards the first category, JAERI entered into an agreement with UKAEA in December 1965 for the exchange of information and co-operation concerning the fast breeder reactor. A similar link with Euratom for co-operation concerning the exchange of information including that on the fast breeder reactor is scheduled to come into effect before long. For international co-operation in the second category, Atomic Power Development Associates of the U.S.A. are inviting Japanese establishments to join their mixed oxide programme, and this is now being considered.

Before I conclude, I should like to mention something about the design

study. At present, the primary concern of the design group is the design of an experimental reactor. As this is scheduled to be almost complete by 1968, the design of the demonstration reactor will be planned after that, and then the design policy for the prototype reactor will be decided. These will be followed by the design of the prototype based on all the information obtained from experiments with the experimental reactor, critical facility and engineering study, and also on the information derived from international co-operation. In implementing the development plan, the procurement of plutonium is considered to be important in the execution of the programme. Japan's first pilot fuel reprocessing plant with a throughput of 200 tons per year is scheduled to start working in 1971. Therefore, the use of Japanese produced plutonium is envisaged for the prototype reactor and later reactors, and it is hoped to use imported plutonium for the critical facility and the experimental reactor. The latter may be operated with enriched uranium as the initial charge fuel, if it is difficult to obtain sufficient plutonium.

The research and development of the first stage are to be conducted as a combined endeavour of all those concerned, from JAERI, AFC, private industry, and universities. The funds required for the activities during this period are roughly estimated at 100 thousand million yen (about £100 million).